

Exercițiul 1.

(3 puncte)

Pentru $z \in \mathbb{C}$, notăm $z = x + iy$, $x, y \in \mathbb{R}$ și avem identificarea $z \simeq (x, y)$. Considerăm funcția $u : \mathbb{C} \rightarrow \mathbb{R}$, $u(x, y) = x^4 + y^4 - 6x^2y^2 - x^3y + xy^3$. Găsiți o funcție olomorfă $f : \mathbb{C} \rightarrow \mathbb{C}$ pentru care $\operatorname{Re}(f) = u$.

Exercițiul 2.

(3 puncte)

Folosind teorema reziduurilor, calculați următoarea integrală:

$$\int_{-\infty}^{\infty} \frac{\cos x}{x^4 + 1} dx.$$

Exercițiul 3.

(3 puncte)

Considerăm $\Omega = \mathbb{C} \setminus \left(\overline{D}\left(\frac{1}{2}, 1\right) \cup \overline{D}\left(-\frac{1}{2}, 1\right) \right)$. Găsiți un șir de aplicații conforme bijective care transformă Ω în semiplanul superior H^+ .

Exercițiul 4.

(3 puncte)

Fie $f : \mathbb{C} \rightarrow \mathbb{C}$ olomorfă, astfel încât imaginea cercului $|z - a| = r$ este inclusă în $\mathbb{R}$. Demonstrați că f este constantă.

Indicație: Metoda 1: Demonstrăm mai întâi că orice funcție armonică neconstantă își atinge atât maximumul cât și minimumul pe frontieră. Folosim acest lucru pentru a demonstra că dacă $f = u + iv$, atunci $v = 0$ pe discul $|z - a| < r$. Deducem apoi că f este constantă.

Metoda 2: Din continuitate, $f(\text{compact}) = \text{compact}$. Folosind teorema aplicației deschise, demonstrăm că pe interiorul cercului (mulțime deschisă), f nu poate lua valori în afara axei reale, decât în cazul în care este constantă.

EXERCITIUL 1

$$z \in \mathbb{C}$$

$$z = x + iy, x, y \in \mathbb{R} ; z = z(x, y)$$

$$+ u: \mathbb{C} \rightarrow \mathbb{R}, u(x, y) = x^4 + y^4 - 6x^2y^2 - x^3y + xy^3$$

$$f: \mathbb{C} \rightarrow \mathbb{C}, f = ? \text{ astfel încât } \operatorname{Re}(f) = u$$

$$\textcircled{\text{Sol}} \frac{\partial u}{\partial x} = 4x^3 - 12xy^2 + y^3 - 3x^2y$$

$$\frac{\partial^2 u}{\partial x^2} = 12x^2 - 12y^2 - 6xy$$

$$\frac{\partial u}{\partial y} = 4y^3 - 12x^2y - x^3 + 3xy^2$$

$$\frac{\partial^2 u}{\partial y^2} = 12y^2 - 12x^2 + 6xy$$

$$\text{Avem c\^a } \Delta u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \Rightarrow u \text{ e armonica}$$

u e armonica

$\mathbb{C}$ e domeniul simpluconex

$$\operatorname{Re}(f) = u$$

$$\Rightarrow \exists f \in H(\mathbb{C}) \text{ a.i. } \operatorname{Re}(f) = u$$

①

Scriem relatia Cauchy-Riemann

$$\begin{cases} \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \\ \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} \end{cases}$$

$$f'(z) = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} = \frac{\partial u}{\partial x} - \frac{\partial u}{\partial y} i$$

$$= 4x^3 - 12xy^2 + y^3 - 3x^2y - (4y^3 - 12x^2y - x^3 + 3xy^2)i$$

Followim teorema de identitate a functiilor omonorfe
luam $y=0$

$$\Rightarrow f'(z) = f'(x) = 4x^3 + x^3i$$

$$\int 4x^3 + x^3i dx = \int 4x^3 dx + \int x^3i dx = x^4 + \frac{i}{4}x^4$$

$$\text{Fie } F: \mathbb{C} \rightarrow \mathbb{C}, f(z) = x^4 + \frac{i}{4}x^4$$

$$\text{Avem ca: } \begin{cases} F'(z) = f'(z), \forall z \in \mathbb{C} \\ \mathbb{R} \text{ are puncte de acumulare in } \mathbb{C} \\ F' \text{ si } f' \text{ sunt omonorfe pe } \mathbb{C} \end{cases}$$

the identity
 $\Rightarrow$
a function
omomorf

$$F'(z) = f'(z), \forall z \in \mathbb{C} \Rightarrow \exists c \in \mathbb{C} \text{ a.i. } F(z) = f(z) + c$$

$$f(z) = F(z) - c \Rightarrow x^4 + \frac{i}{4}x^4 - c = f(z)$$

(2)

$$(x+yi)^4 + \frac{i}{4} (x+yi)^4 - c = f(z)$$

$$(x+yi)^2 = x^2 + 2xyi - y^2$$

$$\Rightarrow (x+yi)^4 = (x^2 + 2xyi - y^2)(x^2 + 2xyi - y^2) =$$

$$x^4 + 2x^3yi - x^2y^2 + 2x^3yi + 4x^2y^2 - 2xy^3i$$

$$- y^2x^2 - 2xy^3i + y^4 =$$

$$= x^4 + 4x^3yi - 6x^2y^2 - 4xy^3i + y^4$$

$$\Rightarrow x^4 + 4x^3yi - 6x^2y^2 - 4xy^3i + y^4 + \frac{i}{4} (x^4 + 4x^3yi - 6x^2y^2$$

$$- 4xy^3i + y^4) - c = f(z) = \frac{i}{4} u(z)$$

$$x^4 + y^4 - 6x^2y^2 + x^3y + xy^3 + \frac{i}{4} (4x^3y - 4xy^3 + x^4 - 6x^2y^2$$

$$+ y^4) - c = f(z) \Rightarrow c = \alpha i, \alpha \in \mathbb{R}$$

③

EXERCITIU 2 $\int_{-\infty}^{\infty} \frac{\cos x}{x^4+1} dx = ?$

Sol $\rightarrow$
 luăm $\int_{-\infty}^{\infty} \frac{\cos x}{x^4+1} dx = I$

$$\int_{-\infty}^{\infty} \frac{\sin x}{x^4+1} dx = J$$

$$I + iJ = \int_{-\infty}^{\infty} \frac{\cos x}{x^4+1} + i \frac{\sin x}{x^4+1} dx = \int_{-\infty}^{\infty} \frac{e^{ix}}{x^4+1} dx$$

$$z^4+1=0 \Rightarrow z^4=-1 = \cos \pi + i \sin \pi$$

$$|z|=1$$

$$z_k = \sqrt[4]{1} \left(\cos \frac{2k\pi + \pi}{4} + i \sin \frac{2k\pi + \pi}{4} \right)$$

$$z_0 = \cos \frac{\pi}{4} + i \sin \frac{\pi}{4} = \frac{\sqrt{2}}{2} + i \frac{\sqrt{2}}{2}$$

$$z_1 = \cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4} = -\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} = -\frac{\sqrt{2}}{2} + i \frac{\sqrt{2}}{2}$$

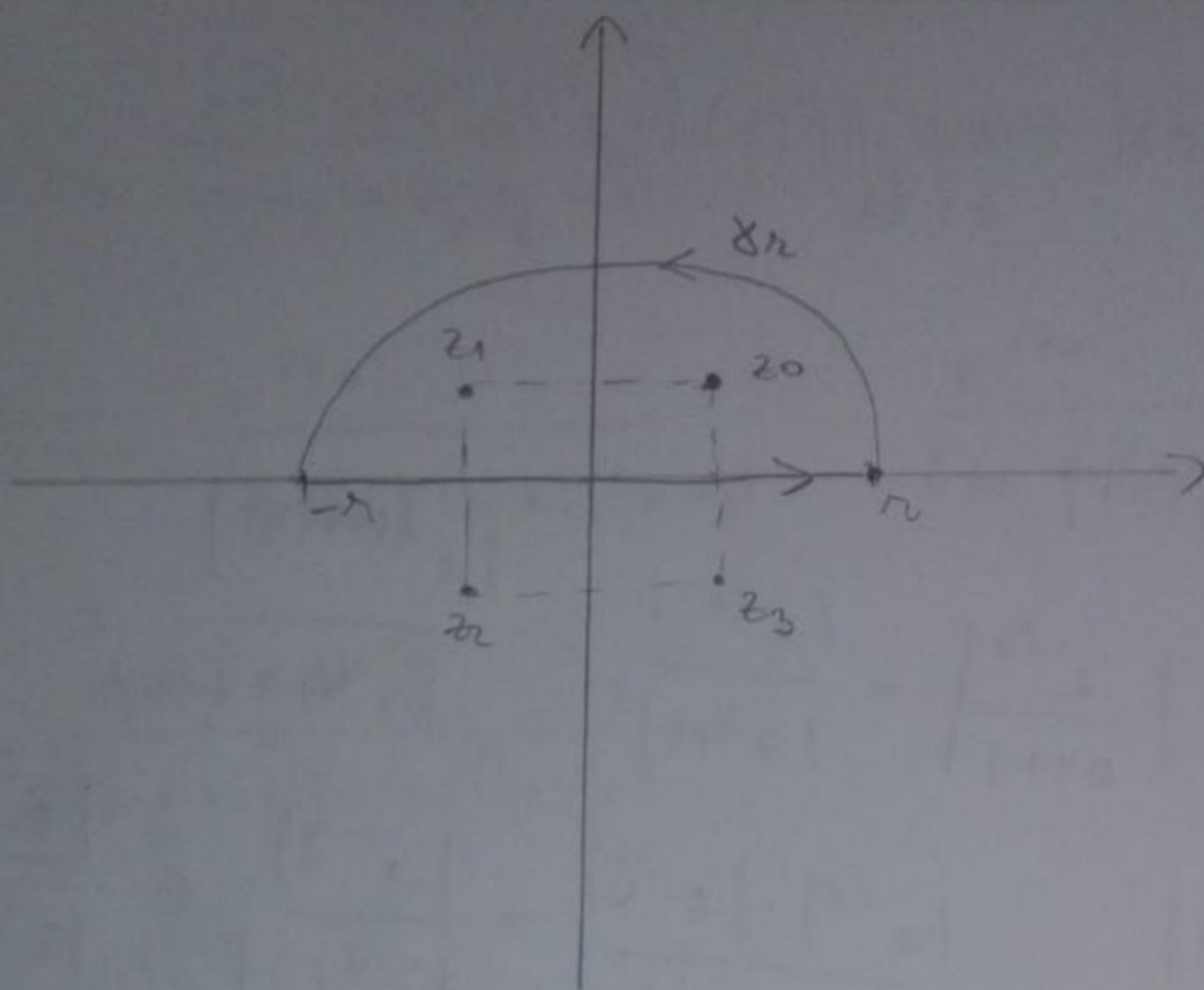
$$z_2 = \cos \frac{5\pi}{4} + i \sin \frac{5\pi}{4} = -\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2} i$$

$$z_3 = \frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2} i$$

für $f: \mathbb{C} \setminus \{z_0, z_1, z_2, z_3\} \rightarrow \mathbb{C}$

$$f(z) = \frac{e^{iz}}{z^4+1}$$

(4)



$$\Gamma_r = [-r, r] \cup \gamma_r, \text{ unde } \gamma_r: [0, \pi] \rightarrow \mathbb{C}, \gamma_r(t) = e^{it}$$

Γ_r e ^{druu} închis, de clasă C_1 pe porțiuni } $\xrightarrow{\text{teorema}}$ reziduurilor
 $\text{Im } \Gamma_r \leq \Delta$

$$\Gamma_r \approx 0$$

$$\int_{\Gamma_r} f(z) dz = 2\pi i \cdot \left(i(\Gamma_r, z_0) \cdot \text{Res } f(z_0) + i(\Gamma_r, z_1) \cdot \text{Res } f(z_1) + \right. \\ \left. i(\Gamma_r, z_2) \cdot \text{Res } f(z_2) + i(\Gamma_r, z_3) \cdot \text{Res } f(z_3) \right)$$

$$i(\Gamma_r, z_0) = 1$$

$$i(\Gamma_r, z_3) = 0$$

$$i(\Gamma_r, z_1) = 1$$

$$i(\Gamma_r, z_2) = 0$$

$$\int_{\Gamma_r} f(z) dz = \underbrace{\int_{-r}^r f(z) dz}_{(5)} + \int_{\gamma_r} f(z) dz$$

$$\left| \int_{\delta r} f(z) dz \right| \leq \sup_{z \in \delta r} |f(z)| \cdot \underbrace{l(\delta r)}_{\pi \cdot r}$$

$$\sup_{z \in \delta r} |f(z)| = ?$$

$$|f(z)| = \left| \frac{e^{iz}}{z^{4+1}} \right| = \frac{|e^{iz}|}{|z^{4+1}|} = \frac{|e^{i(x+iy)}|}{|z^{4+1}|} =$$

$$\frac{|e^{ix-y}|}{|z^{4+1}|} \leq \frac{|e^{ix}| \cdot |e^{-y}|}{|z^{4+1}|} = \frac{|e^{-y}|}{|z^{4+1}|} \leq \frac{|e^{-y}|}{|z^4| - 1}$$

$$\leq \frac{1}{|z^4| - 1} = \frac{1}{r^4 - 1}$$

$$\Rightarrow \left| \int_{\delta r} f(z) dz \right| \leq \frac{1}{r^4 - 1} \cdot \pi r \xrightarrow{r \rightarrow \infty} 0$$

$$\int_{\Gamma_r} f(z) dz = 2\pi i (\text{Res } f(z_0) + \text{Res } f(z_1))$$

$$\text{Res } f(z_0) = \lim_{z \rightarrow z_0} \frac{1}{0!} [(z - z_0)^0 \cdot f(z)]$$

$$= \lim_{z \rightarrow z_0} (z - z_0) \cdot f(z) = \lim_{z \rightarrow z_0} \frac{e^{iz}}{z^{4+1}} \cdot (z - z_0) =$$

(6)

$$\lim_{z \rightarrow z_0} \frac{e^{iz}(z-z_0)}{(z-z_0)(z-z_1)(z-z_2)(z-z_3)} =$$

$$\lim_{z \rightarrow z_0} \frac{e^{iz}}{(z-z_1)(z-z_2)(z-z_3)} = \frac{e^{iz_0}}{(z_0-z_1)(z_0-z_2)(z_0-z_3)}$$

$$= \frac{e^{i(\frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2})}}{(z_0-z_1)(z_0-z_2)(z_0-z_3)}$$

$$= \frac{e^{i(\frac{\sqrt{2}}{2} + \frac{i\sqrt{2}}{2})}}{(\frac{\sqrt{2}}{2} + \frac{i\sqrt{2}}{2} + \frac{\sqrt{2}}{2} - \frac{i\sqrt{2}}{2})(\frac{\sqrt{2}}{2} + \frac{i\sqrt{2}}{2} + \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i)(\frac{\sqrt{2}}{2} + \frac{i\sqrt{2}}{2} - \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i)}$$

$$= \frac{e^{-\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i}}{\sqrt{2} \cdot (\sqrt{2} + i\sqrt{2}) \sqrt{2}i} = \frac{e^{-\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i}}{2i(\sqrt{2} + i\sqrt{2})}$$

$$\frac{e^{-\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i}}{-2\sqrt{2} + 2\sqrt{2}i}$$

$$\text{Res } f(z_1) = \lim_{z \rightarrow z_1} \frac{1}{0!} [(z-z_1) \cdot f(z)] =$$

$$\lim_{z \rightarrow z_1} (z-z_1) \cdot f(z) = \lim_{z \rightarrow z_1} \frac{e^{iz}}{(z-z_0)(z-z_2)(z-z_3)}$$

(7)

$$z \rightarrow \lim_{z \rightarrow z_1} \frac{e^{iz_1}}{(z_1 - z_0)(z_1 - z_2)(z_1 - z_3)^2}$$

$$\frac{e^{i(-\frac{\sqrt{2}}{2} + \frac{i\sqrt{2}}{2})}}{(-\frac{\sqrt{2}}{2} + \frac{i\sqrt{2}}{2} - \frac{\sqrt{2}}{2} - \frac{i\sqrt{2}}{2})(-\frac{\sqrt{2}}{2} + \frac{i\sqrt{2}}{2} + \frac{\sqrt{2}}{2} + \frac{i\sqrt{2}}{2}i) \cdot (-\frac{\sqrt{2}}{2} + \frac{i\sqrt{2}}{2} - \frac{\sqrt{2}}{2} + \frac{i\sqrt{2}}{2}i)}$$

$$z \rightarrow \frac{e^{-\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}i}}{(-\sqrt{2}) \cdot (\sqrt{2}i) \cdot (-\sqrt{2} + i\sqrt{2})} \quad z \rightarrow \frac{e^{-\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}i}}{-2i(-\sqrt{2} + i\sqrt{2})}$$

$$\frac{e^{-\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}i}}{}$$

$$= 2\sqrt{2} + 2\sqrt{2}i$$

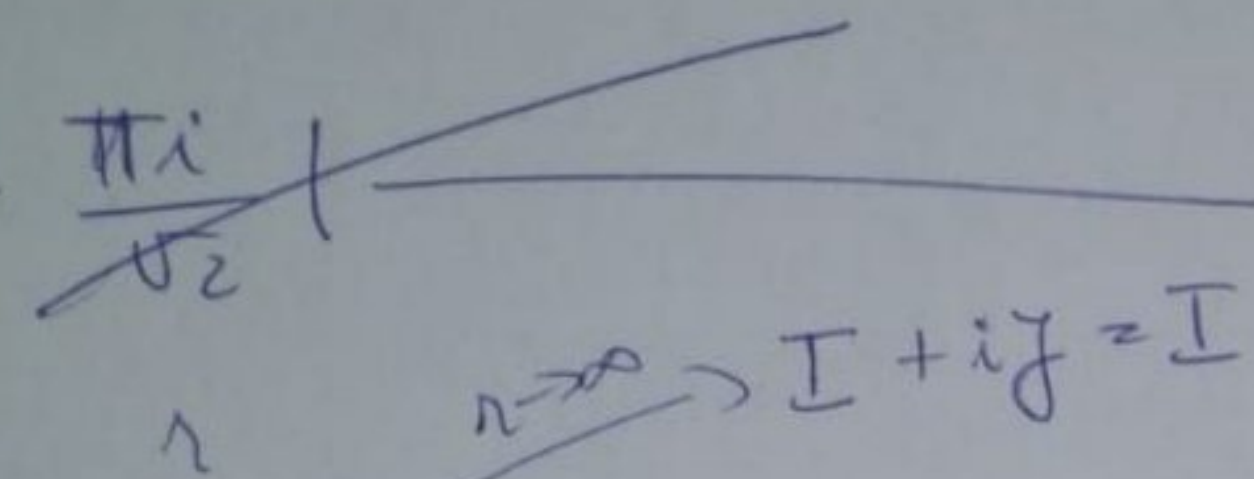
$$z \rightarrow \int_{\Gamma_R} f(z) dz = 2\pi i \left(\frac{e^{-\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i}}{-2\sqrt{2} + 2\sqrt{2}i} + \frac{e^{-\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}i}}{2\sqrt{2} + 2\sqrt{2}i} \right)$$

$$= \frac{2\pi i}{2\sqrt{2}} \left(\frac{e^{-\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i}}{-1+i} + \frac{e^{-\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}i}}{1+i} \right) =$$

$$= \frac{4i}{\sqrt{2}} \left(\frac{(-1-i) \cdot \frac{1}{2} e^{-\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} i} + (1-i) e^{-\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2} i}}{2} \right)$$

dar $\int_{-\infty}^{\infty} \frac{\sin x}{x^4+1} dx = 0$, deoarece $\frac{\sin x}{x^4+1}$ e impară

$\Rightarrow I = \frac{\pi i}{\sqrt{2}}$



$\xrightarrow{r \rightarrow 0} I + i\gamma = I$

$$\int_{\Gamma} f(z) dz = \int_{-R}^R f(z) dz + \int_R^{\infty} f(z) dz \xrightarrow{R \rightarrow \infty} 0$$

u

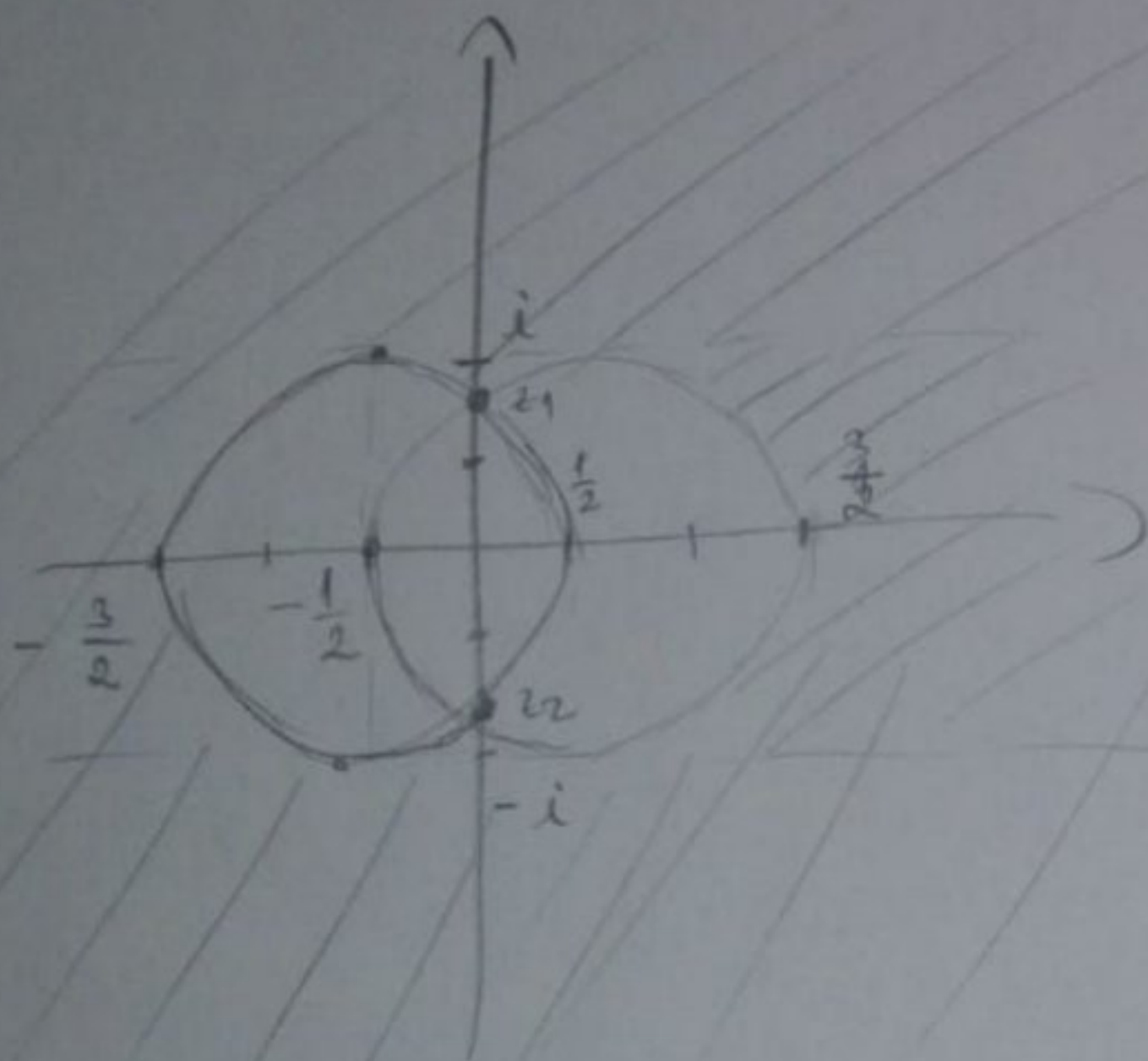
$$2\pi i (\operatorname{Res} f(z_0) + \operatorname{Res} f(z_1))$$

$$\Rightarrow I = \frac{\pi i}{\sqrt{2}} \left((-1-i) \cdot e^{-\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} i} + (1-i) \cdot e^{-\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2} i} \right)$$

Sub 3

$$\Omega = \mathbb{C} \setminus (\overline{D}(\frac{1}{2}, 1) \cup \overline{D}(-\frac{1}{2}, 1))$$

$$\Omega \rightarrow \mathbb{H}^+$$



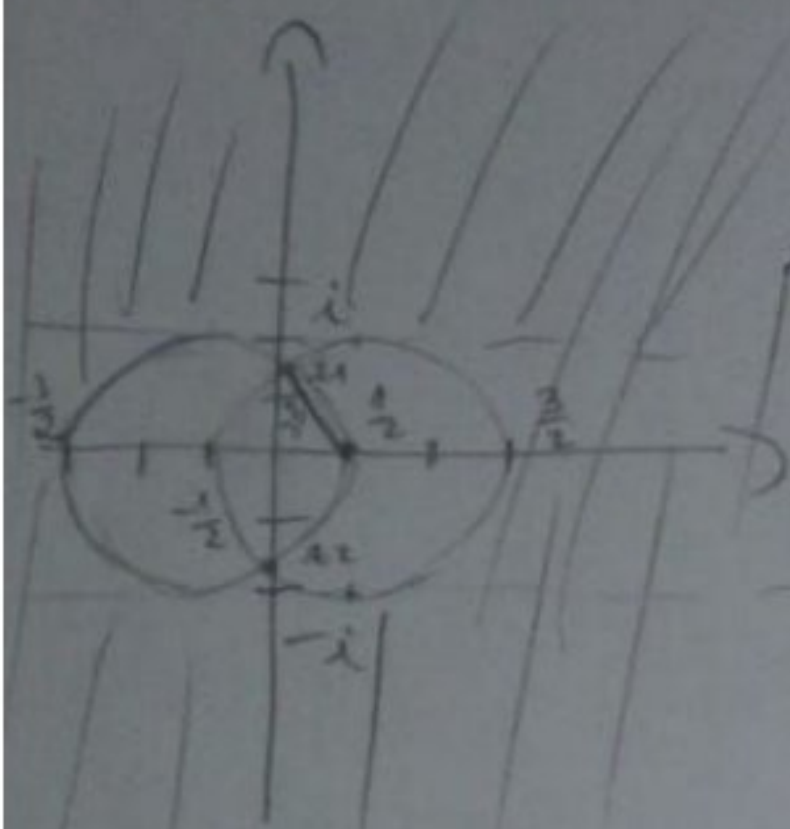
$$\begin{cases} (x - \frac{1}{2})^2 + y^2 = 1 \\ (x + \frac{1}{2})^2 + y^2 = 1 \end{cases} \Rightarrow (x - \frac{1}{2})^2 + y^2 = (x + \frac{1}{2})^2 + y^2$$

$$\Rightarrow x^2 - x + \frac{1}{4} = x^2 + x + \frac{1}{4} \Rightarrow x = 0$$

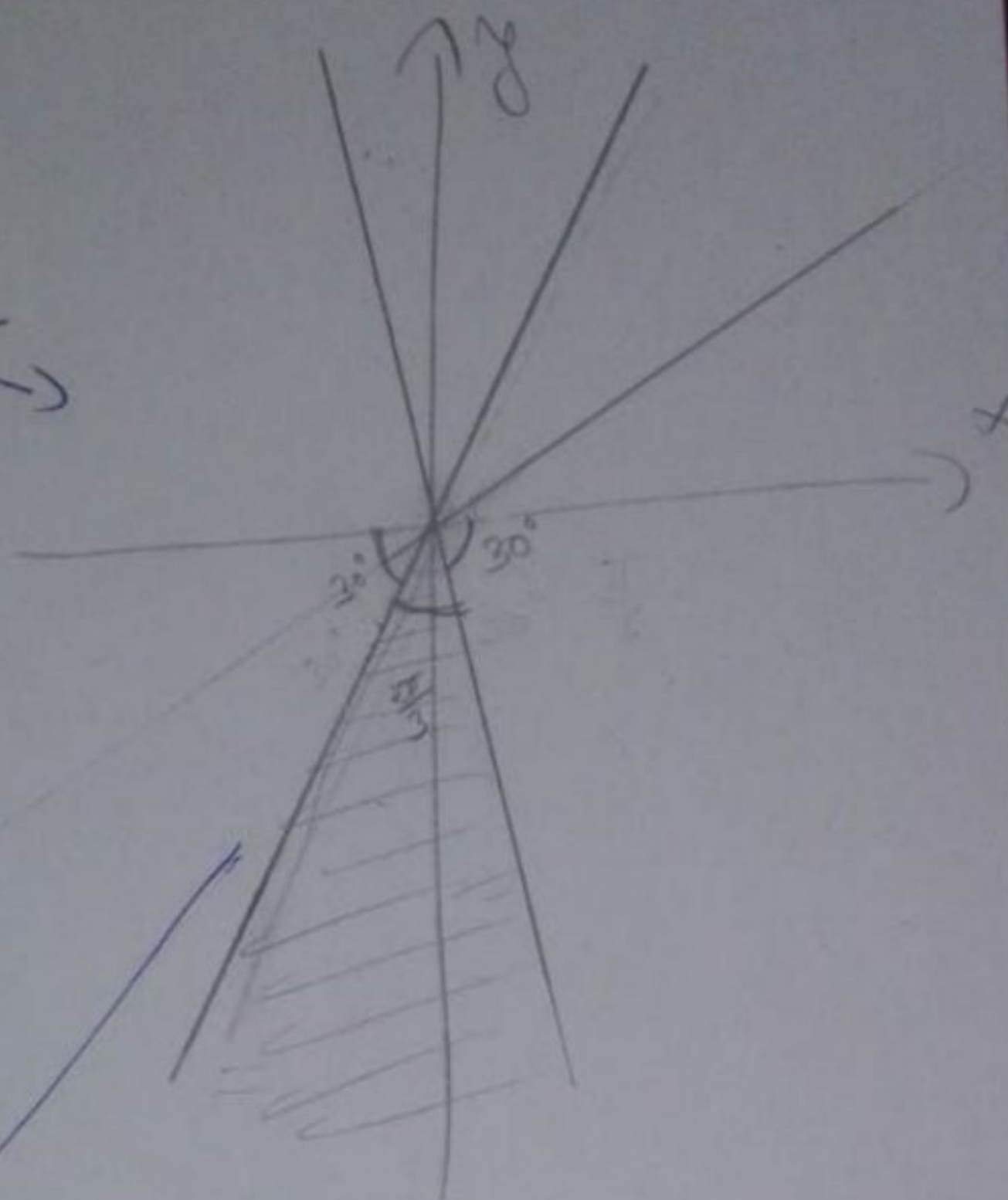
$$\Rightarrow \text{~~no intersection~~} \quad (0 - \frac{1}{2})^2 + y^2 = 1 \Rightarrow$$

$$y^2 = 1 - \frac{1}{4} = \frac{3}{4} \Rightarrow y = \pm \frac{\sqrt{3}}{2} \Rightarrow \begin{aligned} z_1 &= i\frac{\sqrt{3}}{2} \\ z_2 &= -i\frac{\sqrt{3}}{2} \end{aligned}$$

(10)



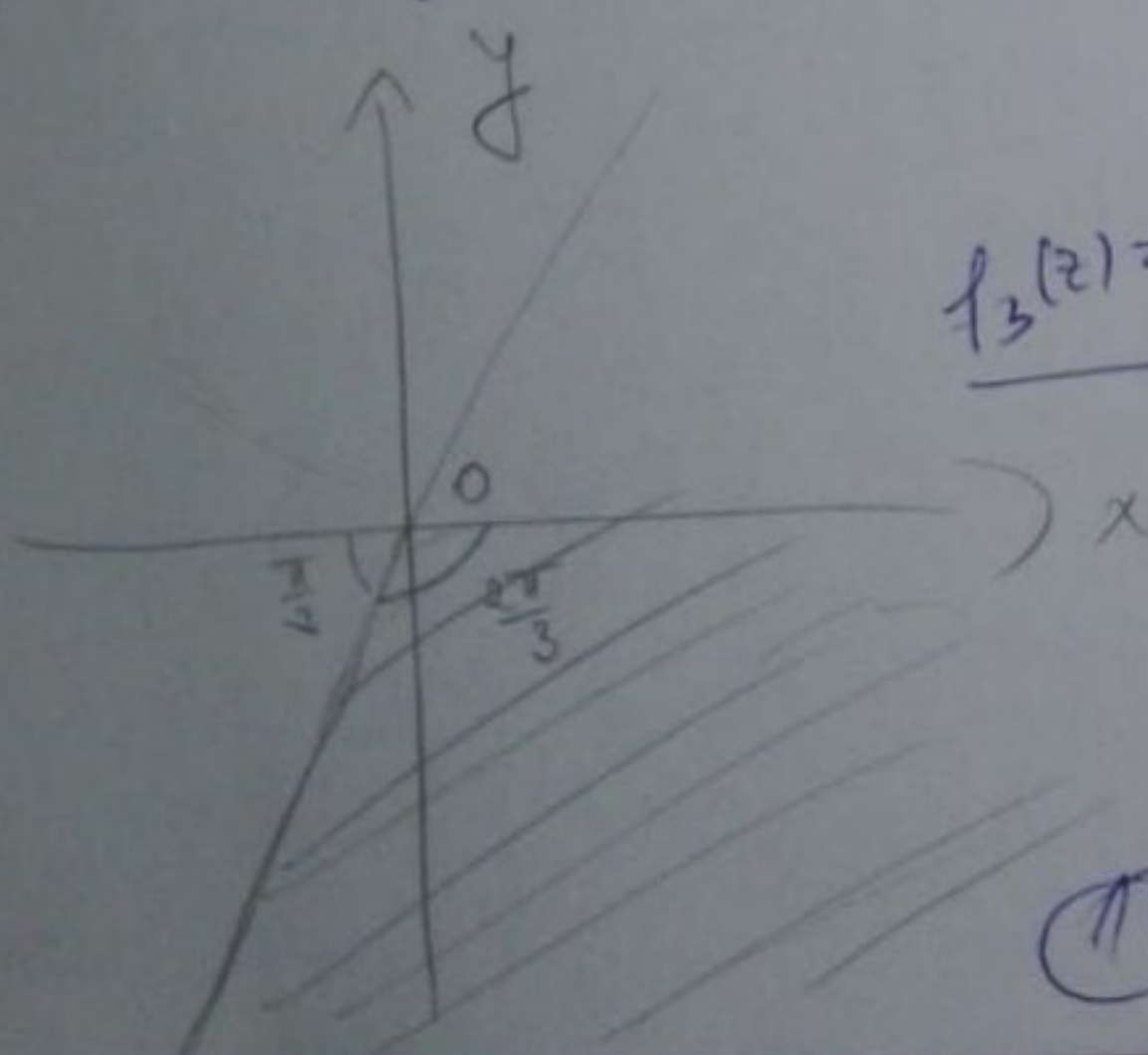
$$f_1(z) = \frac{z - z_1}{z - z_2}$$



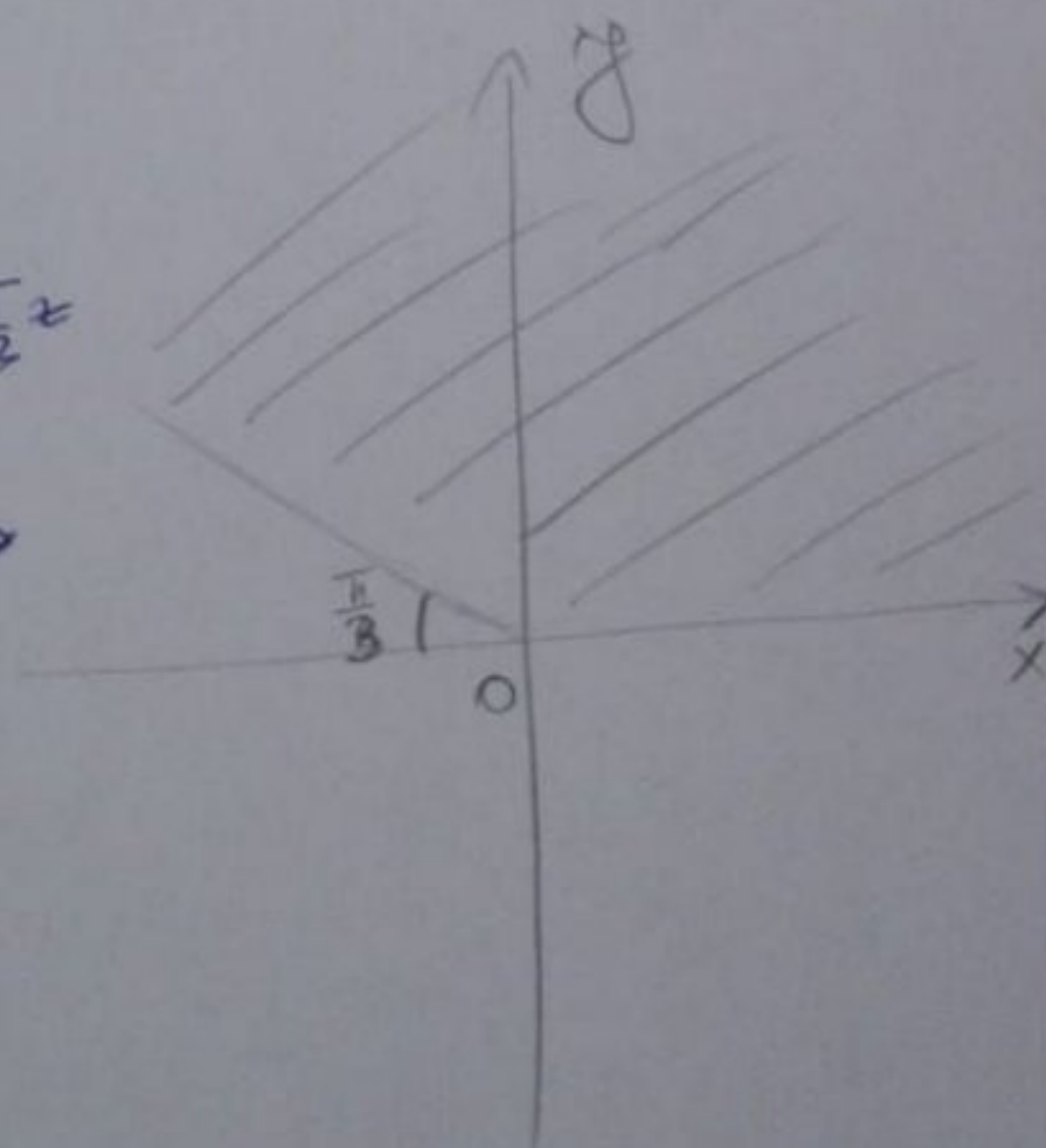
$$f(z) = \frac{z - i\frac{\sqrt{3}}{2}}{z + i\frac{\sqrt{3}}{2}} =$$

$$= \frac{4 - \frac{3}{4} - 2i\frac{\sqrt{3}}{2}}{4 + \frac{3}{4}} = \frac{13 - i\sqrt{3}}{19}$$

$$f_2(z) = e^{i\frac{\pi}{6}z}$$

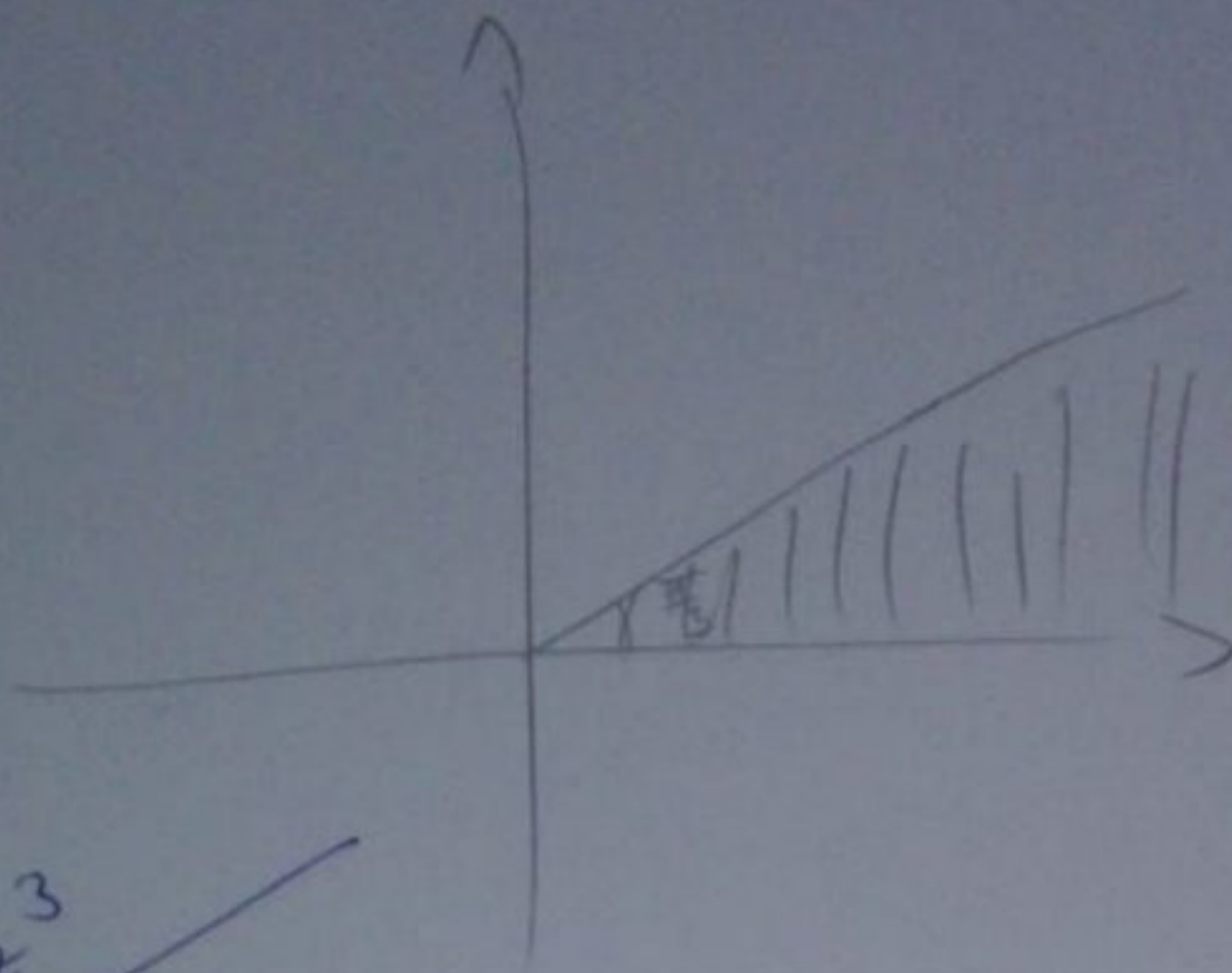


$$f_3(z) = e^{i\frac{2\pi}{3}z}$$

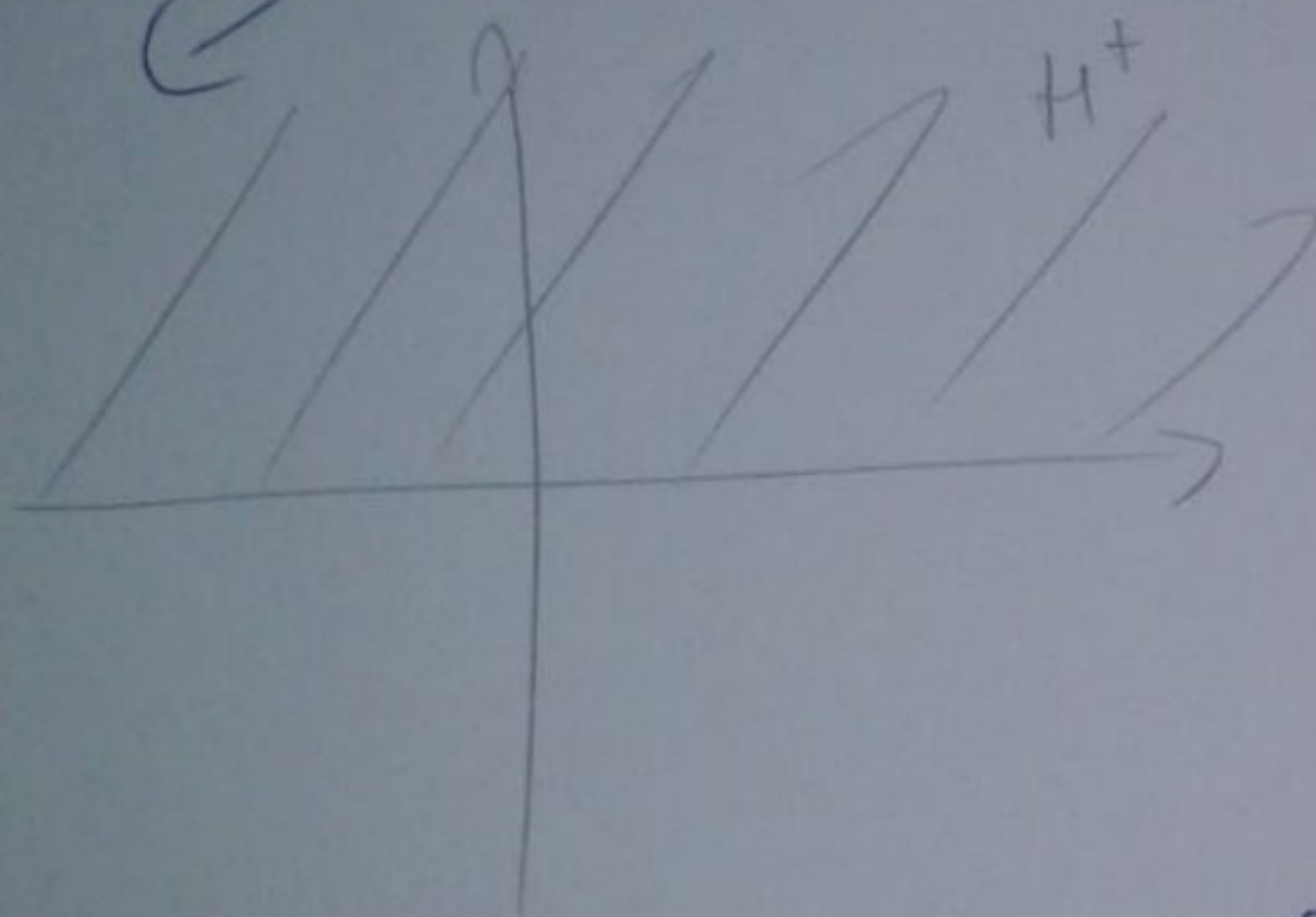


(11)

$$f_4(z) = z^{\frac{1}{2}}$$



$$f_5(z) = z^3$$



f_1, f_2, f_3, f_4, f_5 olomorf $\Rightarrow f = f_5 \circ f_4 \circ f_3 \circ f_2 \circ f_1$ olomorf

$$f = f_5 \circ f_4 \circ f_3 \circ f_2 \circ f_1(z)$$

$$f_2(f_1(z)) = f_2\left(\frac{z-z_1}{z-z_2}\right) = f_2\left(\frac{z-i\frac{\sqrt{3}}{2}}{z+i\frac{\sqrt{3}}{2}}\right)$$

(12)